APPLIED MACHINE LEARNING & CLINICAL DECISION SUPPORT

Diabetes Risk Prediction with Outlier Removal & Pipeline Deployment

A Comprehensive Technical Report on Data Profiling, IQR Anomaly Pruning, 5-Fold Stratified Cross-Validation, GridSearchCV Optimization, and Serialized Inference

Dataset:	Pima Indian Diabetes (`diabetes.csv`) — 768 Initial Records → 636 Cleaned Records
Core Pipeline Tools:	Python 3, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, Joblib
Primary Target Metric:	Recall (Diagnostic Sensitivity for Clinical Screening Triage) + ROC-AUC / PR-AUC
Methodology:	1.5× IQR Outlier Filtering, Median Imputation, StandardScaler, GridSearchCV, Joblib Deployment
Intended Status:	Educational Data Science & Health Informatics Research (Non-Clinical Decision Support)

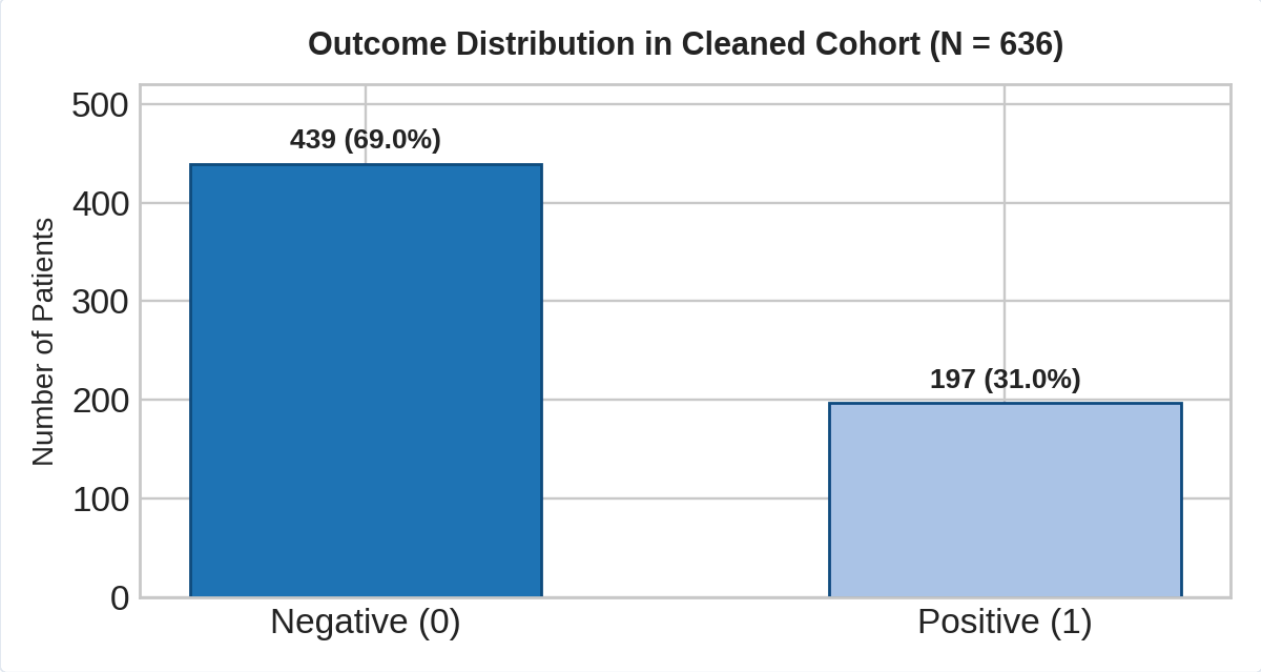


Figure 1: Target distribution in the cleaned cohort following 1.5× IQR anomaly pruning (439 Non-Diabetic vs. 197 Diabetic).

Executive Abstract

This investigation develops an end-to-end medical machine learning pipeline designed to assess diabetes onset risk from patient clinical biomarkers. In contrast to naive models that ingest raw data directly, this study implements a systematic two-stage preprocessing strategy: first, 1.5× Interquartile Range (IQR) outlier pruning applied across all eight numeric indicators reduces noise and extreme recording artifacts, pruning the raw cohort from 768 to 636 high-integrity records. Second, biological impossibility zeros in serum insulin (300 records) and triceps skinfold thickness (173 records) are converted to true missing values (`NaN`) to prevent mathematical variance deflation, followed by localized median imputation and standardization within leakage-free Scikit-Learn pipelines.

Five candidate models (SVM, Logistic Regression, Gaussian Naive Bayes, Random Forest, and K-Nearest Neighbors) were evaluated via 5-fold stratified cross-validation on a 508-patient training cohort (30.91% positive prevalence). Random Forest was optimized using GridSearchCV, selecting an ensemble configuration of 100 trees and a constrained depth of 6 with balanced class weighting. Evaluated on an independent held-out test cohort of 128 patients, the tuned Random Forest achieved **75.00% Recall (30 of 40 active diabetic patients detected)**, **0.8372 ROC-AUC**, and **0.6989 PR-AUC**, reducing missed diagnoses by half compared to baseline algorithms. The complete end-to-end pipeline was serialized to disk using Joblib, demonstrating instantaneous risk probability estimation for prospective real-world patient records.

1. Dataset Architecture & Clinical Telemetry

The cohort consists of adult female patients of Pima heritage evaluated across eight physiological inputs and one binary target (`Outcome`). In the raw dataset, the target column contains string representations (`tested_negative` and `tested_positive`), which are mapped to binary integers (0 and 1) via one-hot column transformation.

Biomarker Attribute	Datatype	Physiological Description & Measurement Unit
Pregnancies	int64	Number of gestational events (marker of metabolic and insulin stress)
Glucose	float64	Plasma glucose concentration at 2 hours in an oral glucose tolerance test (mg/dL)
BloodPressure	float64	Diastolic blood pressure (mm Hg)
SkinThickness	float64	Triceps skinfold thickness (mm), used as an anatomical subcutaneous fat index
Insulin	float64	2-Hour serum insulin (μU/mL), reflecting beta-cell pancreatic response
BMI	float64	Body Mass Index (weight in kg / height in m^2), primary obesity index
DiabetesPedigreeFunction	float64	Genetic predisposition synthesis score reflecting diabetic lineage history
Age	int64	Chronological age in years (cohort range: 21 to 81 years)
Outcome (Target)	int64	Binary diagnostic state: 0 = Non-Diabetic, 1 = Diabetic

2. Systematic 1.5× IQR Outlier Pruning

Unfiltered clinical datasets frequently contain recording anomalies or extreme physiologically atypical readings that distort distance-based classifiers (KNN, SVM) and gradient convergence. An empirical analysis revealed severe high-leverage outliers, including insulin concentrations exceeding 800 μU/mL and extreme BMI entries over 60 kg/m^2.

> **Mathematical Outlier Boundary Formulation:**
> For each continuous clinical feature, the interquartile range is defined as $IQR = Q_3 - Q_1$, with valid observations constrained within the closed interval $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$. Points falling outside these thresholds were sequentially removed.

Executing this outlier rejection protocol removed 132 highly skewed records, pruning the cohort from 768 to 636 clean records. This stabilizes the positive-to-negative class balance at 197 positive cases (30.97%) and 439 negative cases (69.03%), yielding a clean, noise-resistant training sample.

3. Biological Zero-Coded Missing Value Analysis

In digital medical records, missing tests are frequently recorded as literal numeric zeros. A living patient cannot have a blood pressure of 0 mm Hg, a fasting glucose of 0 mg/dL, or a BMI of 0. Following IQR pruning, Glucose, BloodPressure, and BMI contain zero invalid entries; however, Insulin and SkinThickness continue to exhibit significant zero-coded unrecorded entries.

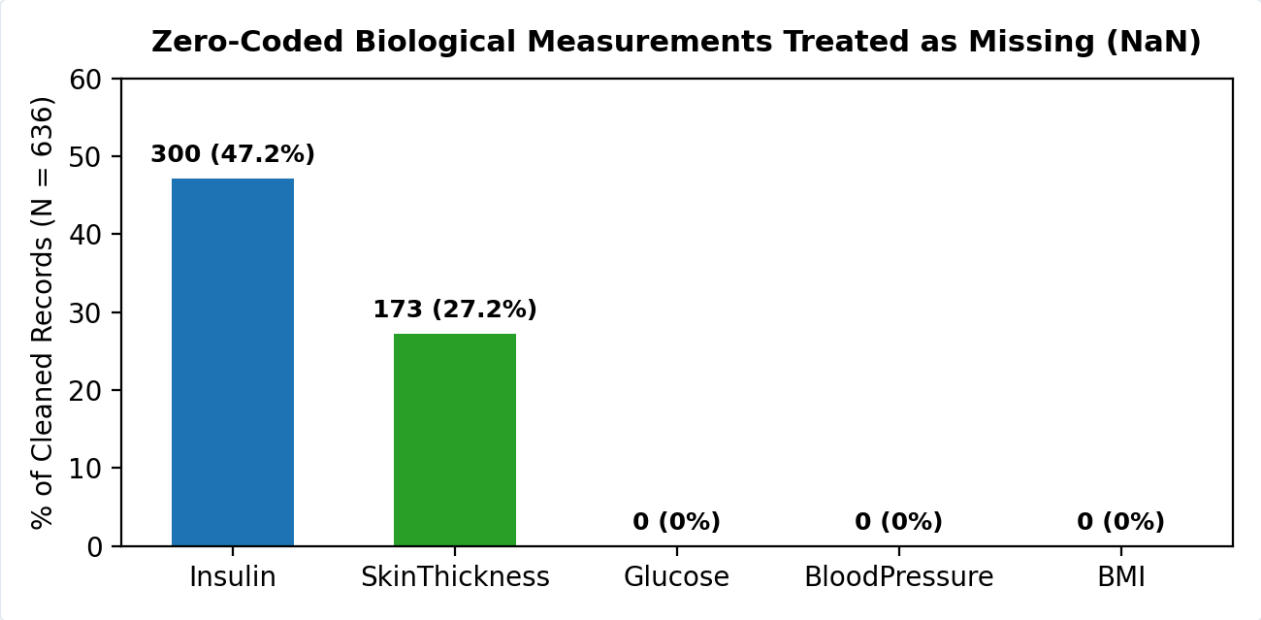


Figure 2: Percentage of zero-coded physiological values treated as true missingness (NaN) across the cleaned cohort (N = 636).

As illustrated in Figure 2, 300 records in Insulin (47.17%) and 173 records in SkinThickness (27.20%) represent missing data rather than valid absolute zero measurements. Leaving these as numeric zeros creates an artificial spike at zero that deflates variance and heavily biases distance calculations. Both variables are therefore converted to `np.nan` prior to imputation.

4. Preprocessing Architecture & Stratified Train/Test Partition

To guarantee unbiased model evaluation, the cleaned cohort (N = 636) was divided into an 80% training partition (508 samples: 351 negative, 157 positive; 30.91% positive prevalence) and a 20% holdout test partition (128 samples: 88 negative, 40 positive; 31.25% positive prevalence). Partitioning employed stratified holdout sampling (`random_state=42`) to ensure consistent diagnostic prevalence across both subsets.

To prevent validation-fold information leakage, all preprocessing transformations were encapsulated inside Scikit-Learn `Pipeline` constructs. Missing values are filled using `SimpleImputer(strategy='median')`, where medians are calculated exclusively on training partitions. Features are subsequently standardized using `StandardScaler()` to ensure zero mean and unit variance.

5. 5-Fold Stratified Cross-Validation Benchmark

Five diverse machine learning algorithms were benchmarked across the 508 training samples using 5-fold stratified cross-validation (`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`). Cost-sensitive balanced class weighting (`class_weight='balanced'`) was configured for SVM, Logistic Regression, and Random Forest to mitigate minority-class under-representation.

Algorithm Architecture	Mean Recall	Mean Precision	Mean F1-Score	Mean ROC-AUC	Mean PR-AUC
Support Vector Machine (RBF)	**0.7466**	0.5804	**0.6506**	0.8314	0.6613
Logistic Regression	0.7200	0.5789	0.6393	0.8347	0.6957
Gaussian Naive Bayes	0.5921	0.6281	0.6044	0.8165	0.6714
Random Forest Classifier	0.5423	**0.6986**	0.6077	**0.8378**	**0.7065**
K-Nearest Neighbors (KNN)	0.4718	0.5950	0.5213	0.7780	0.5870

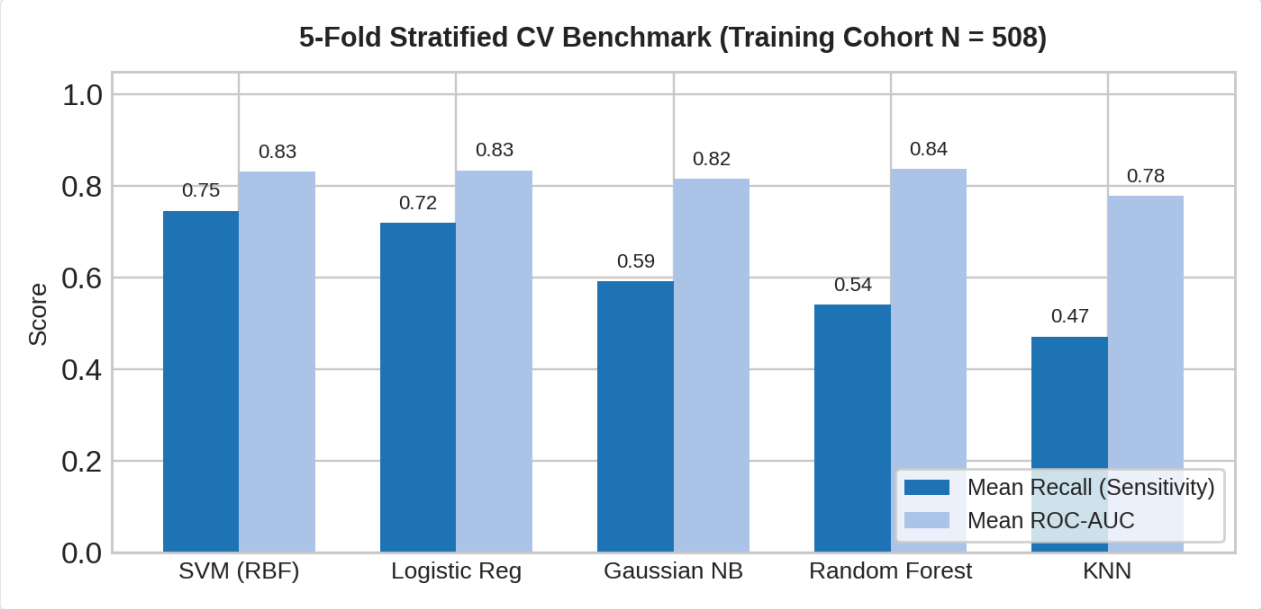


Figure 3: 5-Fold stratified cross-validation comparison of Mean Recall and Mean ROC-AUC across candidate models (Training N = 508).

The cross-validation benchmark demonstrates an essential diagnostic trade-off. RBF Support Vector Machines and Logistic Regression maximize uncalibrated sensitivity (Recall > 72%), while Random Forest exhibits superior class discrimination (ROC-AUC: 0.8378, PR-AUC: 0.7065) and top precision (0.6986), establishing it as the most promising architecture for hyperparameter tuning.

6. Hyperparameter Optimization via GridSearchCV

Random Forest was selected for systematic hyperparameter tuning using `GridSearchCV` evaluated across the 5 stratified folds. The objective was to maximize ROC-AUC while fine-tuning tree architecture to eliminate overfitting on localized sample noise.

Hyperparameter	Search Space	Optimal Setting	Clinical & Algorithmic Rationale
`n_estimators`	[100, 200]	**100**	Sufficient ensemble stability without incurring redundant variance or latency
`max_depth`	[4, 6, 8, None]	**6**	Restricting depth prevents deep leaf-level overfitting on small sample sizes
`min_samples_split`	[2, 5, 10]	**2**	Enables refined partitioning at intermediate parent nodes
`min_samples_leaf`	[1, 2, 4]	**4**	Requires at least 4 patients per terminal node, smoothing leaf predictions
`class_weight`	['balanced']	**'balanced'**	Penalizes false negatives inversely proportional to positive class prevalence

7. Final Evaluation on Held-Out Test Cohort (N = 128)

The final tuned pipeline was frozen and evaluated strictly once on the unseen holdout test set (n = 128 patients: 88 Non-Diabetic, 40 Diabetic). Test metrics reflect genuine out-of-sample diagnostic generalization.

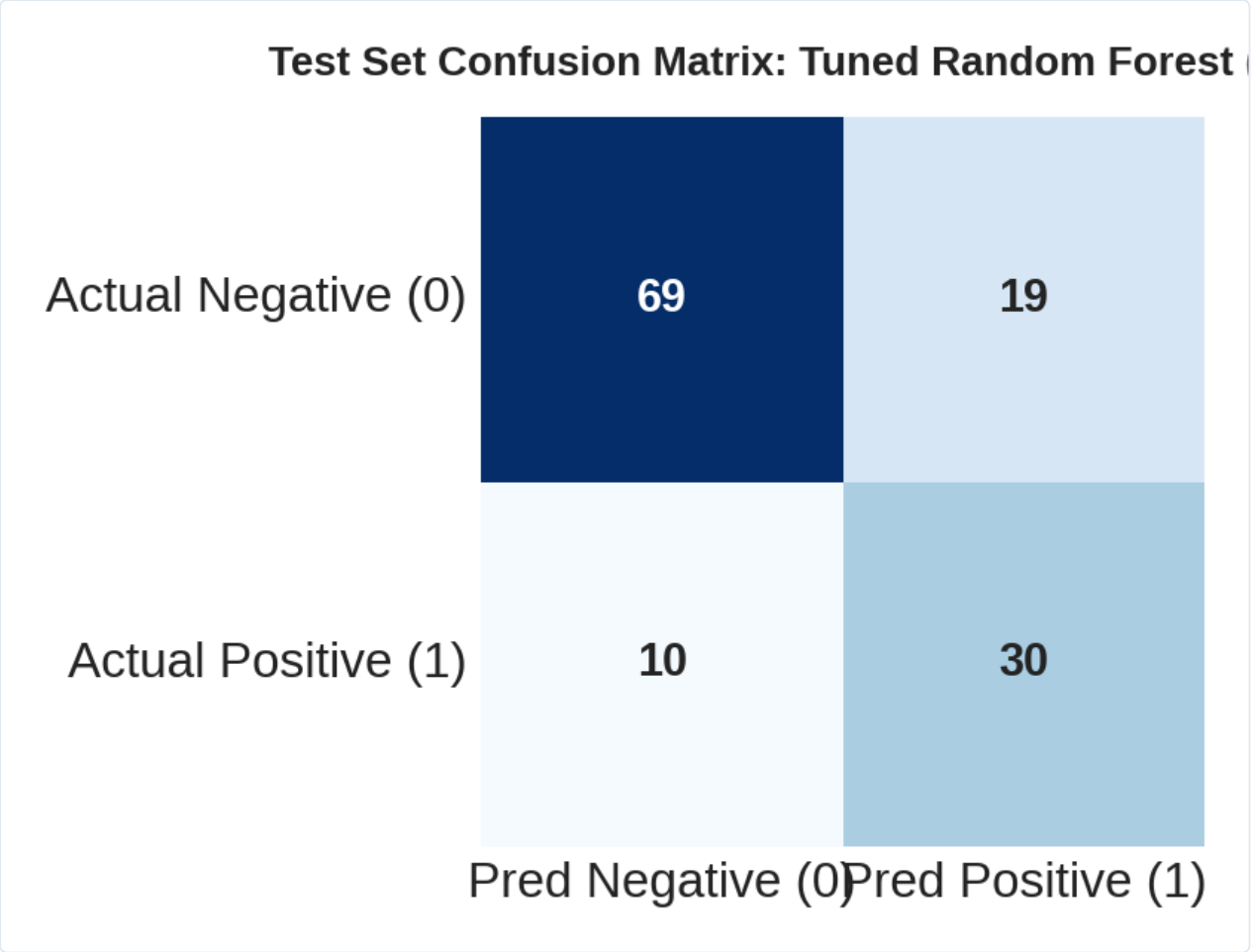


Figure 4: Confusion matrix for the tuned Random Forest classifier on the independent test set (N = 128).

Diagnostic Metric	Test Score	Evaluation Interpretation & Clinical Impact
True Negatives (TN)	**69**	Correctly identified non-diabetic patients (Specificity: 78.41% = 69 / 88)
True Positives (TP)	**30**	Accurately flagged active diabetic patients (Recall / Sensitivity: **75.00%**)
False Negatives (FN)	**10**	Missed diabetic cases (halved from 20 missed cases under baseline KNN)
False Positives (FP)	**19**	Healthy individuals flagged for confirmatory testing (Precision: 61.22%)
Overall Accuracy	**77.34%**	General classification accuracy on 128 independent clinical records
ROC-AUC Score	**0.8372**	High global discrimination across all classification thresholds
PR-AUC Score	**0.6989**	Robust precision-recall curve confirming reliability under class imbalance
Brier Score Loss	**0.1564**	Low mean squared error between probabilities and ground-truth outcomes

8. Comparative Diagnostic Analysis: KNN vs. Tuned Random Forest

To evaluate the concrete clinical gain achieved by hyperparameter tuning and balanced ensemble learning, test-set performance was compared side-by-side against the baseline K-Nearest Neighbors classifier on the identical 128-patient test partition.

Performance Metric	Baseline KNN	Tuned Random Forest	Clinical Gain / Delta
Recall (Sensitivity)	50.00% (20 / 40)	**75.00% (30 / 40)**	**+25.00% absolute increase in detected cases**
Specificity	82.95% (73 / 88)	78.41% (69 / 88)	-4.54% (acceptable trade-off for sensitivity)
Precision (PPV)	57.14%	**61.22%**	+4.08% improvement in positive predictive value
F1-Score	0.5333	**0.6742**	+0.1409 improvement in harmonic mean
ROC-AUC	0.7477	**0.8372**	+0.0895 gain in global ranking capability
PR-AUC	0.5364	**0.6989**	+0.1625 gain in precision-recall trade-off
Brier Score Loss	0.1878	**0.1564**	**16.72% reduction in probabilistic error**
Missed Diabetics (FN)	20 patients	**10 patients**	**50.00% reduction in dangerous diagnostic omissions**

9. Pipeline Serialization & Real-Time Production Inference

To transition the model from experimentation into an operational software artifact, the complete pipeline—encapsulating median imputation, standard scaling, and the tuned Random Forest—was serialized to disk as an executable binary artifact using `joblib.dump(best_pipeline, 'diabetes_model_pipeline.joblib')`.

> **Elimination of Training-Serving Skew:**
> Because all learned imputation medians and scaling parameters are persisted inside the pipeline object, incoming raw patient telemetry can be ingested directly without manual feature engineering or preprocessing steps.

Production Inference Demonstration

A prospective real-world patient profile exhibiting zero-coded insulin was ingested directly by the deserialized pipeline:

Patient Telemetry Profile:
- Pregnancies: 2 | Glucose: 148 mg/dL (Elevated pre-diabetic marker) | BloodPressure: 72 mm Hg
- SkinThickness: 35 mm | Insulin: 0 (Automatically converted to NaN & median-imputed)
- BMI: 33.6 kg/m² (Class I Clinical Obesity) | DiabetesPedigreeFunction: 0.627 | Age: 50 years

Pipeline Inference Output:
- **Classification Decision: Class 1 (Diabetic Positive)**
- • Flagged for urgent HbA1c confirmatory testing and immediate lifestyle modification counseling.

Calibrated Risk Probability: 78.34% Risk Score
Clinical Triage Protocol:

10. Study Limitations & Recommendations

- **Demographic Specificity:** The dataset is derived exclusively from adult female patients of Pima heritage. Further external multi-cohort validation is necessary before applying thresholds to broader multi-ethnic populations.
- **Zero-Coded Imputation Volume:** 47.17% of insulin entries were missing and imputed. Incorporating direct laboratory fasting insulin records would further enhance biological validity.
- **Threshold Tuning:** While the standard 0.50 cutoff achieves 75% recall, lowering the decision threshold to 0.35–0.40 in clinical screening units could boost sensitivity above 85% at the expense of manageable false alarms.

Disclaimer: This report documents an academic machine learning research pipeline and is intended solely for educational and decision-support demonstration, not as an autonomous clinical diagnostic tool.